

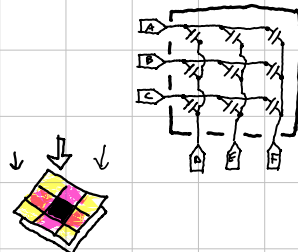
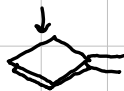
# Ball Plate Balancer

## Option A

- camera w/ opencv to determine position   01011101

## Option B

- capacitive pressure grid
- multiplexed RC circuit measurement
- position from cell with highest capacitance

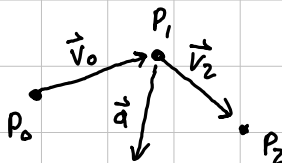


## Estimation of time derivatives of position

→ given 3 positions at 3 times

$$\vec{V} = \frac{\Delta \vec{r}}{\Delta t} \quad \vec{V}_0 = \frac{\vec{P}_1 - \vec{P}_0}{t_1 - t_0}, \quad \vec{V}_1 = \frac{\vec{P}_2 - \vec{P}_1}{t_2 - t_1}$$

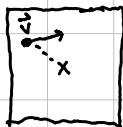
$$\vec{a} = \frac{\Delta \vec{V}}{\Delta t} \Rightarrow \vec{a} = \frac{\vec{V}_1 - \vec{V}_0}{t_2 - t_1}$$



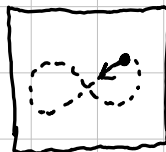
The ball knows where it is because it knows where it isn't. By subtracting where it is from where it isn't, it obtains a difference. - missile

## Steering behavior

- try to "steer" ball back to target



- return to center
- follow some path
- trace sand?

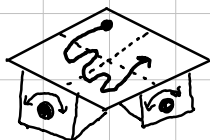


## working principle

$$C = \frac{\epsilon_0 A}{d} \quad kx = mg$$

$$m = \frac{\epsilon_0 A k}{g C}$$

## ball movement mechanism



- Stewart plat form
- ↳ inverse kinematics?
- marble maze game?

